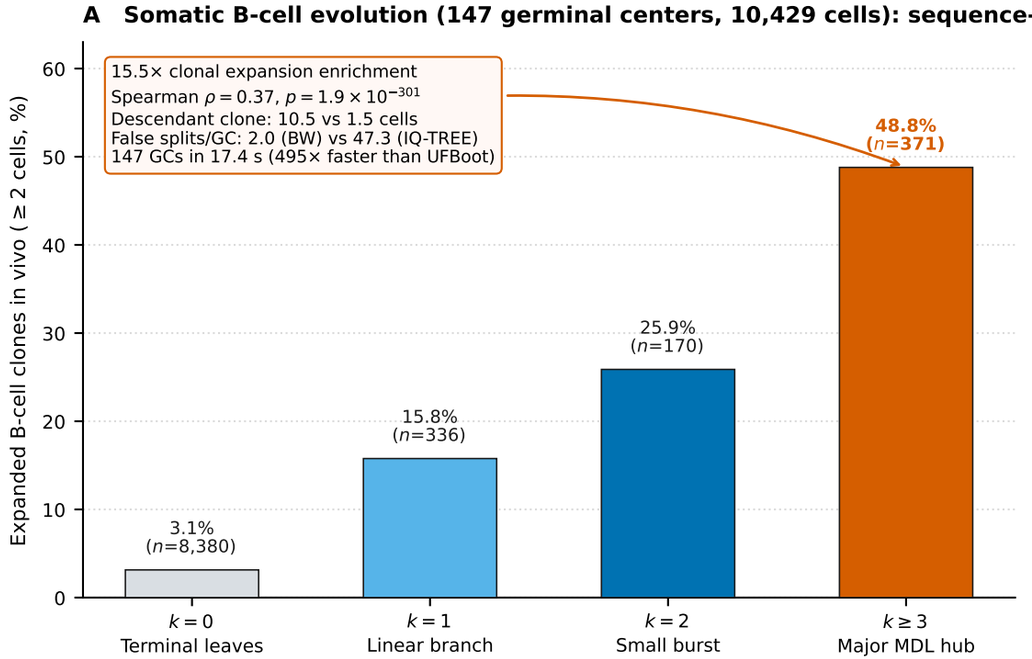
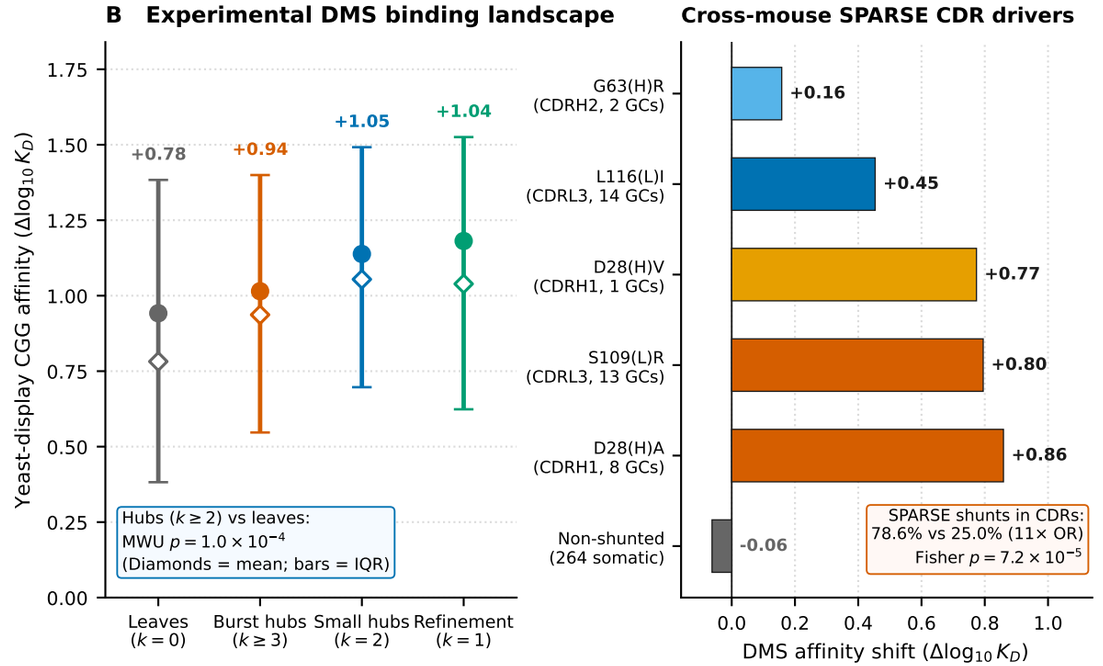
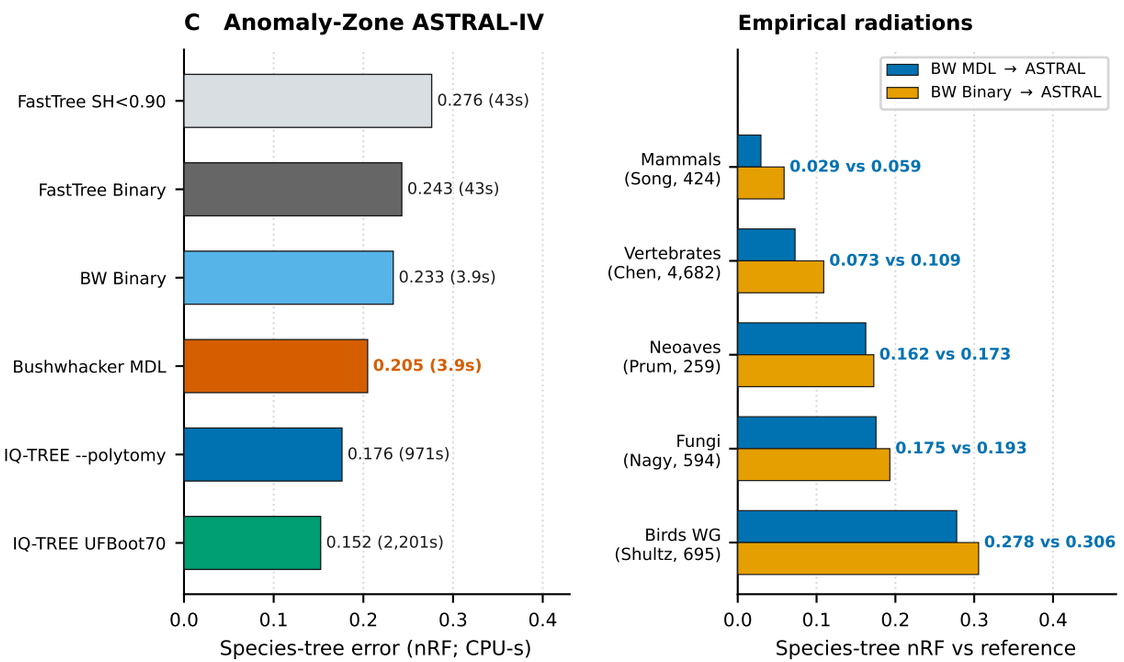
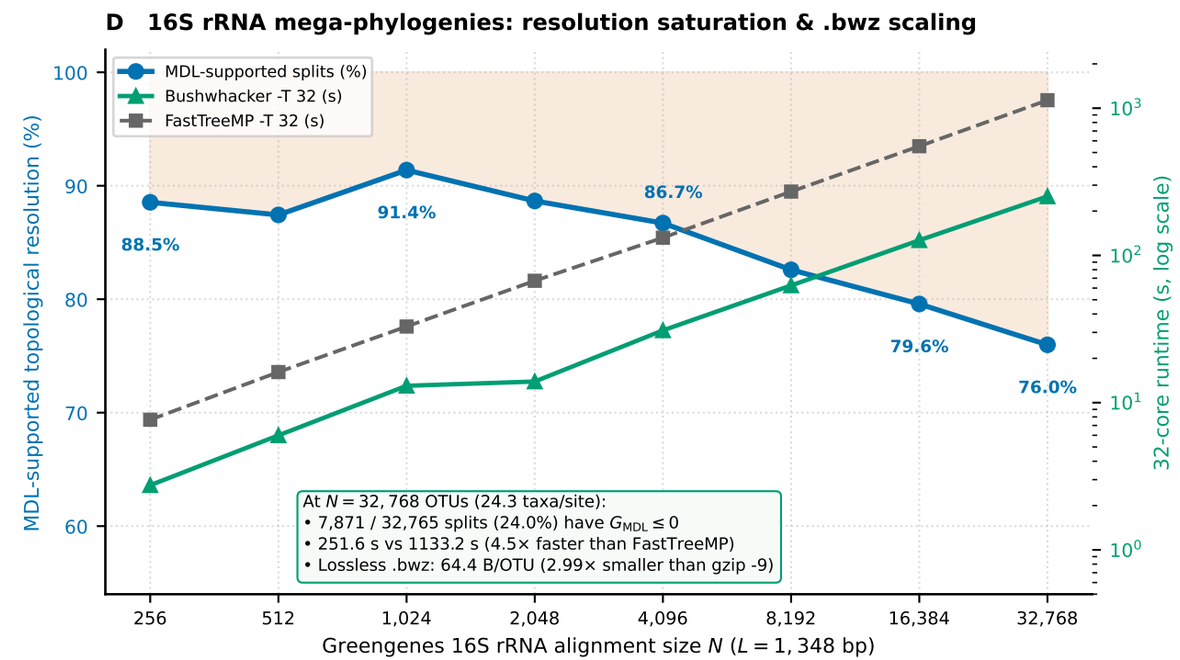

A Somatic B-cell evolution (147 germinal centers, 10,429 cells): sequence-only hubs
15.5× clonal expansion enrichment
Spearman $\rho = 0.37$, $p = 1.9 \times 10^{-301}$
Descendant clone: 10.5 vs 1.5 cells
False splits/GC: 2.0 (BW) vs 47.3 (IQ-TREE)
147 GCs in 17.4 s (495× faster than UFBoot)
Expanded B-cell clones in vivo (≥ 2 cells, %)
3.1% (n=8,380)
15.8% (n=336)
25.9% (n=170)
48.8% (n=371)
k = 0 Terminal leaves
k = 1 Linear branch
k = 2 Small burst
k ≥ 3 Major MDL hub


B Experimental DMS binding landscape
Yeast-display CGG affinity (Δlog10 KD)
+0.78
+0.94
+1.05
+1.04
Hubs (k ≥ 2) vs leaves:
MWU p = 1.0 × 10−4
(Diamonds = mean; bars = IQR)
Leaves (k = 0)
Burst hubs (k ≥ 3)
Small hubs (k = 2)
Refinement (k = 1)
Cross-mouse SPARSE CDR drivers
G63(H)R (CDRH2, 2 GCs)
L116(L)I (CDRL3, 14 GCs)
D28(H)V (CDRH1, 1 GCs)
S109(L)R (CDRL3, 13 GCs)
D28(H)A (CDRH1, 8 GCs)
Non-shunted (264 somatic)
+0.16
+0.45
+0.77
+0.80
+0.86
-0.06
SPARSE shunts in CDRs:
78.6% vs 25.0% (11× OR)
Fisher p = 7.2 × 10−5
DMS affinity shift (Δlog10 KD)


C Anomaly-Zone ASTRAL-IV
FastTree SH<0.90
FastTree Binary
BW Binary
Bushwhacker MDL
IQ-TREE --polytomy
IQ-TREE UFBoot70
0.276 (43s)
0.243 (43s)
0.233 (3.9s)
0.205 (3.9s)
0.176 (971s)
0.152 (2,201s)
Species-tree error (nRF; CPU-s)

Empirical radiations
BW MDL → ASTRAL
BW Binary → ASTRAL
Mammals (Song, 424)
Vertebrates (Chen, 4,682)
Neoaves (Prum, 259)
Fungi (Nagy, 594)
Birds WG (Shultz, 695)
0.029 vs 0.059
0.073 vs 0.109
0.162 vs 0.173
0.175 vs 0.193
0.278 vs 0.306
Species-tree nRF vs reference


D 16S rRNA mega-phylogenies: resolution saturation & .bwz scaling
MDL-supported splits (%)
Bushwhacker -T 32 (s)
FastTreeMP -T 32 (s)
MDL-supported topological resolution (%)
88.5%
91.4%
86.7%
79.6%
76.0%
At N = 32, 768 OTUs (24.3 taxa/site):
• 7,871 / 32,765 splits (24.0%) have GMDL ≤ 0
• 251.6 s vs 1133.2 s (4.5× faster than FastTreeMP)
• Lossless .bwz: 64.4 B/OTU (2.99× smaller than gzip -9)
32-core runtime (s, log scale)
Greengenes 16S rRNA alignment size N (L = 1, 348 bp)